



SANDIP
UNIVERSITY

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Neelam Vidya Vihar, Sijoul, Mailam, Madhubani, Bihar – 847235

Website: <http://www.sandipuniversity.edu.in> Email: info@sandipuniversity.edu.in

Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Algorithm Analysis and Design	Short Title : AAD
Course Code : CSAIML301	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1.	Understand fundamental algorithmic concepts, analysis and mathematical analysis of recursive and non-recursive algorithms.
2.	Apply brute-force and divide-and-conquer strategies to solve problems like sorting, searching, and matrix multiplication.
3.	Design efficient solutions using greedy and dynamic programming approaches for optimization problems.
4.	Implement backtracking and branch-and-bound techniques for constraint satisfaction and combinatorial problems.
5.	Analyze computational complexity, lower bounds, and classify problems into P, NP, and NP-Complete categories.

Course Outcomes :

1.	Analyze algorithms using asymptotic notations and solve recurrence relations for recursive and non-recursive algorithms.
2.	Implement brute-force and divide-and-conquer techniques.
3.	Solve optimization problems using greedy and dynamic programming approaches.
4.	Apply backtracking and branch-and-bound methods to solve combinatorial problems.
5.	Classify computational problems into P, NP, and NP-Complete categories using lower-bound theory.

Unit No.	Details	Hours
1	Basic Concepts of Algorithms: Notion of Algorithm, Fundamentals of Algorithmic Solving, Important problem types, Fundamentals of the Analysis Framework, Asymptotic Notations and Basic Efficiency Classes, Mathematical analysis of non-recursive algorithms. Mathematical analysis of recursive algorithm: recurrence relations, solution of recurrence relations using substitution method	8

Dr. J. K. Singh *Dr. Anurag Singh* *Dr. J. K. Singh*



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2	Brute Force, Divide and Conquer Strategy: Selection sort, Bubble sort, Sequential searching (Linear Search), Brute force string matching, General method, Merge sort, Quick Sort, Binary Search, Strassen's matrix multiplication	8
3	Greedy Approach and Dynamic Programming: Fractional Knapsack problem, Minimum cost spanning tree: Prim's and Kruskal's algorithm, Single source shortest path problem, Principle of optimality, Multi-stage graph problem, all pair shortest path problem, 0/1 Knapsack problem, Traveling salesperson problem.	8
4	Backtracking and Branch and Bound : General method backtracking, N-Queen problem, 0/1 Knapsack problem, General method of branch & bound, 0/1 Knapsack problem, Traveling sales person problem	8
5	Lower Bound Theory and Complexity Classes: Lower bounds, Decision trees, P, NP and NP Complete problems	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. Algorithm Design, Jon Kellinberg and Eva Tardos, 1st Edition, Pearson Education 2014. 2. Design & Analysis of Algorithms, Gajendra Sharma, Khanna Book Publishing 2018. 3. Fundamentals of algorithms, Horowitz E, Sahini S, Rajasekaran S., University Press 2008. 4. Introduction to algorithms, Cormen, Leiserson, Rivest, Stein, 3rd Edition, PHI. 2012 5. An introduction to analysis of algorithms, R. Sedgewick, 1st edition, Pearson Education 1996. 6. Data Structures and Program Design in C By Robert L. Kruse, C.L. Tondo, Bruce Leung, Pearson Education. 2007.
E-Resources	1. http://nptel.ac.in/

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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Database Systems	Short Title : DBS
Course Code : CSAIML302	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :	
1	Understand fundamental database concepts, data models, and classifications of DBMS, including concurrency control techniques.
2	Design and optimize relational databases using normalization, functional dependencies, and query processing for structured and semi-structured data.
3	Implement transaction management with ACID properties, serializability, and recovery mechanisms in database systems.
4	Explore techniques for managing unstructured data, including information retrieval, document ranking, and search systems.
5	Apply Big Data platforms and algorithms to process and analyze large-scale structured and unstructured data.
Course Outcomes :	
1	Explain fundamental database concepts, data models, and classify DBMS types, including concurrency control techniques.
2	Design normalized relational databases using functional dependencies and optimizes queries, while managing semi-structured data.
3	Implement transaction management ensuring ACID properties, serializability, and recoverability in database systems.
4	Process unstructured text using information retrieval techniques for document ranking and search.
5	Apply Big Data platforms and algorithms to manage and analyze large-scale structured/unstructured data and graphs.

Unit No.	Details	Hours
1	Introduction: Characteristics and fundamental concepts of Databases, Types of	08

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	Data Models and Data Modelling, Elements of Database Systems, Classification and comparison of Database Management Systems (Regular and NoSQL Page), concurrency control, Lock based concurrency control, Time stamping methods.	
2	Structured and semi-structured data management: Structured data, relational databases, Relational model, Functional Dependencies, normal forms, algorithms for query optimization, Semi-structured data, document-databases, semi-structured data abstraction, representation, and search.	08
3	Transaction Management: Transaction concept, transaction state, ACID properties, serializability, Recoverability, Testing for serializability.	08
4	Unstructured Data Management: Unstructured text, Information retrieval systems, document retrieval and ranking.	08
5	Big Data Management Platforms for Big Data, algorithms for Map-Reduce & Hadoop, Platforms for Big Graphs, algorithms for large graphs.	08
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. Database System Concepts, Abraham Silberschatz, Henry F. Korth, S. Sudharshan, Tata McGraw Hill, 2006 2. Fundamentals of Database Systems, Elmsari and Navathe, Pearson Education 2013 3. Database Management Systems, Ramakrishnan and Gehrke, McGrawHill 2003 4. "An Introduction to Database Systems", C.J.Date, A.Kannan, S.Swamynathan, Pearson Education, 2006 5. Database Management Systems, R.P. Mahapatra, Khanna Book Publishing 2016. 6. J. D. Ullman, "Principles of Database Systems", 2nd Ed., Galgotia Publications 7. Learning Spark: Lightning-Fast Big Data Analysis / Holden Karau, Andy Konwinski, Patrick Wendell, Matei Zaharia / O'Reilly Media; 1st edition / ISBN-13: 978-1449358624 / ISBN-10: 1449358624 8. Data on the Web: From Relations to Semistructured Data and XML / Serge Abiteboul, Peter Buneman, Dan Suciu / 1st Edition / ISBN-13: 978-1558606227 / 9. Introduction to Information Retrieval / Christopher Manning, Prabhakar Raghavan, Hinrich Schütze / book and slides available online
E-Resources	1. http://nptel.ac.in/

Dish *Prabhakar* *Sudharshan*



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Program: B. Tech in Computer Science and Engineering (AI & ML)

Year : Second Year	Semester : III
Course Title : Computer Networks	Short Title : CN
Course Code : CSAIML303	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1	Explain fundamental Internet architecture, packet-switching concepts, and protocol layering models.
2	Design network applications using application layer protocols and socket programming.
3	Analyze transport layer services, including UDP/TCP protocols, reliable data transfer, and congestion control mechanisms.
4	Evaluate network layer functions, IP addressing/routing algorithms, and router operations in Internetworking.
5	Implement data link layer techniques for error handling, multiple access protocols, and LAN switching.

Course Outcomes :

1	Explain Internet architecture, packet-switching fundamentals, and protocol layer models with their service functions.
2	Develop network applications using application-layer protocols and socket programming interfaces.
3	Implement transport-layer protocols with reliable data transfer and congestion control mechanisms.
4	Configure network-layer components and analyze router operations in datagram networks.
5	Apply data-link layer techniques to design efficient local networks.

Unit No.	Details	Hours
1	Computer Networks and The Internet: What is the Internet; network edge; network core; Delay, Loss and throughput in Packet-Switched Networks; Protocol Layers and their Service Models.	8
2	Application Layer: Principles of Network Applications; The Web and HTTP; File Transfer: FTP;	8

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	Electronic Mail in the Internet; DNS - The Internet's Directory Service; Peer-to-Peer applications; Socket Programming – Creating network applications.	
3	Transport Layer: Introduction and Transport-Layer Services; Multiplexing and Demultiplexing; Connectionless Transport: UDP; Principles of Reliable of Data Transfer; Connection-Oriented Transport: TCP; Principles of Congestion Control, TCP Congestion Control.	8
4	Network Layer: Introduction; Virtual circuit and datagram networks; What is inside a router; Internet Protocol (IP): Forwarding and Addressing in the Internet; Routing Algorithms; Routing in the Internet; Broadcast and Multicast Routing.	8
5	Data Link Layer: Introduction to the link layer; Error Detection and Correction Techniques; Multiple Access links and Protocols; Switched local area networks.	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. James F. Kurose and Keith W. Ross, "Computer Networking: A top-down approach", Pearson Education, 6th edition. 2012 2. A.S. Tanenbaum, "Computer Networks", 5th Edition, PHI 2010 3. Bhavneet Sidhu, "An Integrated Approach to Computer Networks", Khanna Book Publishing House 2019. 4. G. Keiser, "Local Area Networks", 2nd Edition, TMH 2002 5. D. Bertsekas and R. Gallager, "Data Networks", 2nd Edition, PHI 2000 6. William Stallings, "Data & Computer Communication", PHI, 10th Edition 2013 7. B.A. Forouzan, "Data communications and networking", TMH, 5th Edition 2012 8. B.A. Forouzan, "Local Area Networks", TMH. 2002 9. B.A. Forouzan, "TCP/IP Protocol Suite", TMH. 2004
E-Resources	1. http://nptel.ac.in/

Dr. Ashutosh Singh



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Program: B. Tech in Computer Science and Engineering (AI & ML)

Year : Second Year	Semester : III
Course Title : Introduction to Machine Learning	Short Title : IML
Course Code : CSAIML304	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1	Understand fundamental machine learning concepts, learning paradigms, and dataset handling techniques including feature engineering and validation methods.
2	Implement supervised learning algorithms for classification and regression tasks, and evaluate model performance using appropriate metrics.
3	Apply unsupervised learning techniques including clustering algorithms to discover patterns in unlabeled datasets.
4	Utilize dimensionality reduction methods and advanced machine learning models to solve complex data analysis problems.
5	Explore cutting-edge machine learning domains including deep learning, NLP and computer vision applications.

Course Outcomes :

1	Explain fundamental machine learning concepts, dataset handling, and evaluation techniques (cross-validation, feature sets).
2	Implement supervised learning algorithms and evaluate models using metrics.
3	Apply unsupervised learning techniques to analyze unlabeled datasets.
4	Utilize dimensionality reduction methods and advanced models for data preprocessing and pattern recognition.
5	Explore emerging domains and deep learning basics to solve real-world problems.

Unit No.	Details	Hours
1	Introduction: What Is Machine Learning?, How Do We Define Learning?, How Do We Evaluate Our Networks?, How Do We Learn Our Network?, What are datasets and how to handle them?, Feature sets, Dataset division: test, train and validation sets, cross validation.	8
2	Basics of machine learning:	8

Devi Ashutosh Yadav



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	Applications of Machine Learning, processes involved in Machine Learning, Introduction to Machine Learning Techniques: Supervised Learning, Unsupervised Learning and Reinforcement Learning, Real life examples of Machine Learning.	
3	Supervised learning: Classification and Regression: K-Nearest Neighbor, Linear Regression, Logistic Regression, Support Vector Machine (SVM), Evaluation Measures: SSE, MME, R ² , confusion matrix, precision, recall, F-Score, ROC-Curve.	8
4	Unsupervised learning: Introduction to clustering, Types of Clustering: Hierarchical, Agglomerative Clustering and Divisive clustering; Partitional Clustering - K-means clustering.	8
5	Miscellaneous: Dimensionality reduction techniques: PCA, LDA, ICA. Introduction to Deep Learning, Gaussian Mixture Models, Natural Language Processing, Computer Vision.	8
Total (Hrs)		40

Resources :	
Text Books & Reference Books	1. Introduction to Machine Learning, By Jeeva Jose, Khanna Book Publishing Co., 2020. 2. Machine Learning for Dummies, By John Paul Mueller and Luca Massaron, For Dummies, 2016. 3. Machine Learning, By Rajeev Chopra, Khanna Book Publishing Co., 2021. 4. Machine Learning: The New AI, By Ethem Alpaydin, The MIT Press, 2016. 5. Machine Learning, Tom M. Mitchell, McGraw Hill Education, 2017.
E-Resources	1. https://www.udacity.com/course/intro-to-machine-learning--ud120 2. https://www.coursera.org/learn/machine-learning-duke 3. http://nptel.ac.in/

Dr. Ashish Kumar



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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Artificial Intelligence	Short Title : AI
Course Code : CSAIML305	
Theory : 3 Hours/Week	Tutorial: 1 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 4

Course Objectives :

1	Understand fundamental AI concepts, intelligent agents, and real-world applications while analyzing AI's societal impact and career opportunities.
2	Implement state-space search strategies and game-theoretic techniques for problem-solving.
3	Apply propositional/predicate logic with resolution techniques and knowledge representation methods for reasoning tasks.
4	Design planning systems using state-space search, partial-order planning, and multi-agent approaches for complex decision-making.
5	Evaluate AI techniques and their limitations across different application domains.

Course Outcomes :

1	Demonstrate understanding of AI fundamentals including intelligent agents, applications, and societal impact through real-world case studies.
2	Implement heuristic search algorithms and game-playing to solve computational problems.
3	Apply logical reasoning techniques and knowledge representation methods to model AI systems.
4	Design automated planning systems using state-space search, partial-order planning, and multi-agent approaches for complex decision-making scenarios.
5	Evaluate advanced AI reasoning methods and their limitations in different problem domains.

Unit No.	Details	Hours
1	Introduction: Artificial Intelligence and its applications, Artificial Intelligence Techniques, Level of models, criteria of success, Intelligent Agents, Nature of Agents, Learning Agents. AI Techniques, advantages, and limitations of AI, Impact and Examples of AI, Application domains of AI. The AI Ladder - The Journey for Adopting AI Successfully, Advice for a career in AI, Hotbeds of AI Innovation.	8

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2	Problem solving techniques: State space search, control strategies, heuristic search, problem characteristics, production system characteristics., Generate and test, Hill climbing, best first search, A* search, Constraint satisfaction problem, Mean-end analysis, Min-Max Search, Alpha-Beta Pruning, Additional refinements, Iterative Deepening.	8
3	Logic: Propositional logic, predicate logic, Resolution, Resolution in propositional logic and predicate logic, Clause form, unification algorithm,	8
4	Knowledge Representation schemes and reasoning: Mapping between facts and representations, Approaches to knowledge representation, procedural vs declarative knowledge, Forward vs. Backward reasoning, Matching, conflict resolution, Non- monotonic reasoning, Default reasoning, statistical reasoning, fuzzy logic Weak and Strong filler structures, semantic nets, frame, conceptual dependency, scripts.	8
5	Planning: The Planning problem, planning with state space search, partial order planning, planning graphs, planning with propositional logic, Analysis of planning approaches, Hierarchical planning, conditional planning, Continuous and Multi Agent planning.	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. A Classical Approach to Artificial Intelligence, M.C. Trivedi, Khanna Book Publishing, 2019. 2. Artificial Intelligence: A modern approach by Stuart Russel, Pearson Education, 2010. 3. Artificial Intelligence by Rich and Knight, The McGraw Hill, 2017. 4. Artificial Intelligence: A new synthesis by Nils and Nilson, Elsevier, 1997. 5. Artificial Intelligence by Luger, Pearson Education, 2002. 6. Artificial Intelligence by Padhy, Oxford Press, 2005.
E-Resources	1. https://www.udemy.com/course/artificial-intelligence-az/ 2. https://www.edx.org/course/artificial-intelligence-ai 3. http://nptel.ac.in/

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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Internet of Things	Short Title : IoT
Course Code : CSAIML306A	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1	Understand IoT fundamentals, ecosystem architecture, and differentiate between embedded systems and IoT devices.
2	Develop practical IoT solutions using Raspberry Pi/Arduino, including sensor integration and remote access setup.
3	Compare wireless communication protocols for optimal IoT deployment scenarios.
4	Design IoT applications for smart cities, healthcare, and Industry 4.0 using sensors, positioning systems, and data processing techniques.
5	Implement security measures and noise-filtering methods to ensure robust and privacy-aware IoT systems.

Course Outcomes :

1	Explain IoT fundamentals including ecosystem architecture, decision frameworks, and differentiate between embedded vs. IoT devices through comparative analysis.
2	Develop IoT prototypes using Raspberry Pi/Arduino by configuring sensing-actuation systems and remote access tools.
3	Evaluate wireless protocols by comparing their properties for optimal IoT deployment in short/long-range scenarios.
4	Design IoT solutions for real-world applications using appropriate sensors and data processing techniques.
5	Implement security measures and noise-filtering methods for sensor data while addressing privacy concerns in IoT systems.

Dr. Sh *Anurag* *Gadgil*



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Unit No.	Details	Hours
1	Introduction to IoT: What is IoT, how does it work? Difference between Embedded device and IoT device, Properties of IoT device, IoT Ecosystem, IoT Decision Framework, IoT Solution Architecture Models, Major IoT Boards in Market.	8
2	Setting Up Raspberry/Arduino to Create Solutions: Explore Raspberry Pi, setting up Raspberry Pi, showing working of Raspberry Pi using SSH Client and Team Viewer, Understand Sensing actions, Understand Actuators and MEMS	8
3	Communication Protocols used in IoT: Types of wireless communication, Major wireless Short- range communication devices, properties, comparison of these devices (Bluetooth, WIFI, ZigBee, 6LoWPAN), Major wireless Long-range communication devices, properties, comparison of these devices (Cellular IoT, LPWAN)	8
4	IoT Applications: Industrial Internet 4.0, Applications such as: Smart home, wearables, smart city, smart grid, connected car, connected health (digital health, telehealth, telemedicine), smart retail.	8
5	Sensors: Applications of various sensors: Google Maps, Waze, WhatsApp, Ola Positioning sensors: encoders and accelerometers, Image sensors: cameras, Global positioning sensors: GPS, GLONASS, IRNSS, Galileo and indoor localization systems, Motion & Orientation Sensors: Accelerometer, Magnetometer, Proximity Sensor, Gyroscope Calibration, noise modeling and characterization and noise filtering and sensor data processing. Privacy & Security	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. Vijay Madisetti and Arshdeep Bahga, Internet of Things (A Hands-on Approach), 1st Edition, VPT, 2014 2. Francis da Costa, Rethinking the Internet of Things: A Scalable Approach to Connecting Everything, 1st Edition, Apress Publications, 2014 3. Cuno Pfister, Getting Started with the Internet of Things, O Reilly Media, 2011 4. Kyung, C.-M., Yasuura, H., Liu, Y., Lin, Y.-L., Smart Sensors and Systems, Springer International Publishing, 2015
E-Resources	1. http://nptel.ac.in/

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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Robotics	Short Title : Robotics
Course Code : CSAIML306B	
Theory : 3 Hours/Week	Tutorial: 0 Hour/Week
Continuous Internal Assessment: 50 Marks	Maximum University Theory Exam: 50 Marks
Time of Theory Exam: 2.5 hrs	Credit : 3

Course Objectives :

1	Understand fundamental robotics concepts, including kinematics, dynamics, and robot programming languages.
2	Analyze the role of AI in robotics, including intelligent agents, decision-making, and game-playing strategies.
3	Apply kinematic and dynamic techniques for robot motion planning, trajectory control, and transformations.
4	Evaluate real-world robotics applications, including intelligent robots, DDD concepts, and accuracy/repeatability metrics.
5	Apply ML techniques to solve real-world problems across different domains and data types.

Course Outcomes :

1	Apply kinematic and dynamic principles to analyze robot motion, trajectory planning, and control systems.
2	Develop robot programming solutions using hierarchical levels and address language-specific challenges.
3	Design AI-driven strategies for game playing and decision-making in robotics using intelligent agents and evaluation functions.
4	Compare robotic architectures, drive systems and specifications for real-world applications.
5	Implement robotics solutions in domains like industrial automation or intelligent systems using DDD concepts and robot anatomy principles.

Unit No.	Details	Hours
1	Introduction: Introduction to Robotics Fundamentals of Robotics, Robot Kinematics: Position Analysis, Dynamic Analysis and Forces, Robot Programming languages & systems: Introduction, the three levels of robot programming, requirements of a robot programming language, problems peculiar to robot pro-	8

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	gramming languages.	
2	Need of AI in Robotics: History, state of the art, Need for AI in Robotics. Thinking and acting humanly, intelligent agents, structure of agents.	8
3	Game Playing: AI and game playing, plausible move generator, static evaluation move generator, game playing strategies, problems in game playing.	8
4	Robotics fundamentals: Robot Classification, Robot Specification, notation, kinematic representations and transformations, dynamics techniques; trajectory planning and control.	8
5	Robotics and Its applications: DDD concept, Intelligent robots, Robot anatomy-Definition, law of robotics, History and Terminology of Robotics-Accuracy and repeatability of Robotics-Simple problems-Specifications of Robot-Speed of Robot, Robot joints and links-Robot classifications-Architecture of robotic systems-Robot Drive systems-Hydraulic, Pneumatic and Electric system	8
Total (Hrs)		40

Resources :

Text Books & Reference Books	1. Robotics, Vision and Control: Fundamental Algorithms in MATLAB, Peter Corke, Springer, 2011. 2. Robotics: Everything You Need to Know About Robotics from Beginner to Expert, Peter McKinnon, Createspace Independent Publishing Platform, 2016. 3. Introduction to AI Robotics, Second Edition, By Robin R. Murphy, MIT press, 2001. 4. Artificial Intelligence for Robotics: Build intelligent robots that perform human tasks using AI techniques, Francis X. Govers, Packt Publishers, 2018.
E-Resources	1. http://nptel.ac.in/

Disol *Ashutosh* *Gadaw*



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Program: B. Tech in Computer Science and Engineering (AI & ML)

Year : Second Year	Semester : III
Course Title : Analysis of Algorithm and Design Lab	Short Title : AAD Lab
Course Code : CSAIML307	Practical: 2 Hours/Week
Term Work: 25 Marks	ESE: 25 Marks
Time of Practical Exam: 2.5 hrs	Credit: 1

Course Objectives :

1	To analyze time and space complexity of various algorithms.
2	To develop algorithm for solving various computational problems by applying various algorithm design strategies.
3	To understand the effect of choice of data structures on the complexity of algorithm.
4	To design efficient solutions for NP-hard problems using approximation techniques and heuristics.
5	To evaluate real-world applicability of sorting, searching, and graph algorithms based on computational constraints.

Course Outcomes :

1	Apply the best data structure for designing an algorithm to solve a given problem.
2	Evaluate different algorithms with respect to time and space complexity.
3	Create algorithms to solve various computational problems.
4	Understand different complexity classes.
5	Compare graph traversal, sorting, and searching algorithms based on real-world constraints.

Sr. No.	Practical Description
Concern faculty member should suitably frame at least Eight laboratory assignments from the following list.	
1	Write a program to perform Bubble sort for any given list of numbers.
2	Write a program to perform Insertion sort for any given list of numbers.
3	Write a program to perform Quick Sort for the given list of integer values.
4	Write a Program to perform Merge Sort on the given two lists of integer values

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5	Write a Program to perform Binary Search for a given set of integer values recursively and non-recursively.
6	Write a program to find solution for knapsack problem using greedy method.
7	Write a program to find minimum cost spanning tree using Prim's Algorithm
8	Write a program to find minimum cost spanning tree using Kruskal's Algorithm.
9	Write a program to perform Single source shortest path problem for a given graph.
10	Write a program for all pairs shortest path problem
11	Write a program find maximum number of edge disjoint paths between two vertices

Practical/Oral/Presentation:

Practical/Oral/Presentation shall be conducted and assessed jointly by internal and external examiners. The performance in the Practical/Oral/Presentation examination shall be assessed by at least a pair of examiners appointed as examiners by the University. The examiners will prepare the mark/grade sheet in the format as specified by the University.

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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Database Systems Lab	Short Title : DBS Lab
Course Code : CSAIML308	Practical: 2 Hours/Week
Term Work: 25 Marks	ESE: 25 Marks
Time of Practical Exam: 2.5 hrs	Credit: 1

Course Objectives :

1	Design efficient relational database schemas using normalization techniques and ER modeling principles.
2	Implement advanced SQL operations to manipulate and query relational data.
	Develop secure database applications with front-end integration and stored procedures.
3	Evaluate concurrency control methods to maintain ACID properties in multi-user environments.
	Compare SQL and NoSQL systems by implementing solutions for structured and unstructured data scenarios.

Course Outcomes :

1	Design and implement normalized relational databases using ER modeling and SQL DDL commands.
2	Develop complex SQL queries involving joins, subqueries, and aggregation functions for data retrieval and manipulation.
3	Implement transaction management with ACID properties using concurrency control techniques.
4	Build database applications with front-end interfaces and stored procedures.
5	Compare NoSQL and SQL systems by implementing document/column-family databases for specific use cases.

Practical Description	
Concern faculty member should suitably frame at least Eight laboratory assignments from the following list.	
1	Creating a sample database using any client server RDBMS package using SQL DDL queries. This will include constraints.
2	SQL DML queries: Use of SQL DML queries to retrieve, insert, update and delete the database.

Dish

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Godav



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3	SQL Queries: The queries should involve SQL feature such as aggregate functions, group by, having, order by clause.
4	SQL Queries: The queries should involve Set Operations and Set Comparisons on the database.
5	Develop two tier architecture and use ODBC/JDBC connections to store and retrieve data from the database. You may consider any applications like employee management system, library management system etc.
6	Execute DDL statements which demonstrate the use of views. Try to update the base table using its corresponding view.
7	Write a database trigger on Library table. The System should keep track of the records that are being updated or deleted. The old value of updated or deleted records should be added in Library_Audit table.
8	Write a program to demonstrate PL/SQL block
9	Write a program to demonstrate PL/SQL stored procedure.
10	Write a program to demonstrate different types of Join.
11	Write a program to demonstrate PL/SQL stored function

Practical/Oral/Presentation:

Practical/Oral/Presentation shall be conducted and assessed jointly by internal and external examiners. The performance in the Practical/Oral/Presentation examination shall be assessed by at least a pair of examiners appointed as examiners by the University. The examiners will prepare the mark/grade sheet in the format as specified by the University.

Dish *Ashish* *Gadav*



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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Computer Networks Lab	Short Title : CN Lab
Course Code : CSAIML309	Practical: 2 Hours/Week
Term Work: 25 Marks	ESE: 25 Marks
Time of Practical Exam: 2.5 hrs	Credit: 1

Course Objectives :

1	Configure network devices and troubleshoot connectivity issues in simulated environments using industry-standard tools.
2	Implement core networking protocols through hands-on socket programming and packet analysis exercises.
3	Design efficient IP networks by applying subnetting, VLANs, and dynamic routing concepts.
4	Analyze network performance and security vulnerabilities using monitoring and diagnostic tools.
5	Compare wired and wireless network technologies through practical experimentation and measurement.

Course Outcomes :

1	Configure and troubleshoot network topologies using routers, switches, and hubs in simulated environments.
2	Implement core network protocols through socket programming and packet analysis.
3	Demonstrate subnetting, VLANs, and routing algorithms to design functional IP networks.
4	Analyze network performance metrics (latency, throughput) and security threats using monitoring tools.
5	Compare wired and wireless communication protocols through practical experimentation.

Sr. No.	Practical Description
Concern faculty member should suitably frame at least Eight laboratory assignments from the following list.	
1	Simulate packet switching delays using Python's time and queue modules to model router buffers.
2	Capture HTTP requests/responses using Wireshark and analyze headers (GET, POST, status codes).

Dr. Ashish

Ashish

Pradeep



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3	Implement a basic FTP client in Python using ftplib to upload/download files.
4	Design a DNS query tool in Python (dnspython) to resolve domain names and trace iterative vs. recursive queries.
5	Build a peer-to-peer chat application using Python sockets (socket library) with multicast support.
6	Compare UDP vs. TCP performance by transmitting video chunks over both protocols and measuring loss/throughput.
7	Code a TCP-like reliable data transfer protocol (ACKs, retransmissions) over an unreliable UDP channel.
8	Visualize TCP congestion control (slow start, AIMD) using ns-3 or Linux tc commands.
9	Configure static routing tables on Linux (ip route) and trace packet paths with traceroute.
10	Implement Dijkstra's algorithm in Python to compute shortest paths for a toy network topology.
11	Analyze ARP traffic with Wireshark to map IP addresses to MAC addresses on your LAN.
12	Simulate CSMA/CD collisions using a Python script with randomized backoff timers.
13	Set up a VLAN using vconfig on Linux and test inter-VLAN communication.
14	Benchmark Ethernet vs. Wi-Fi throughput using iperf3 under varying signal strengths.
15	Debug a network with intentional errors (CRC mismatches, MTU issues) using ping, tcpdump.

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Dr. Ashutosh *Gadgil*



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Program: B. Tech in Computer Science and Engineering (AI & ML)	
Year : Second Year	Semester : III
Course Title : Introduction to Machine Learning Lab	Short Title : IML Lab
Course Code : CSAIML310	Practical: 2 Hours/Week
Term Work: 25 Marks	ESE: 25 Marks
Time of Practical Exam: 2.5 hrs	Credit: 1

Course Objectives :

1	To Understand the basics of Machine Learning.
2	To learn and apply different machine learning models to various datasets.
3	Master foundational ML algorithms through hands-on implementation in Python/R.
4	Develop skills to rigorously evaluate models using appropriate metrics and validation techniques.
5	Design effective data preprocessing pipelines including feature engineering and dimensionality reduction.

Course Outcomes :

1	Understand basic applications and issues of Machine Learning.
2	Understand the different types of datasets.
3	Analyze and work with different datasets.
4	Analyze various Machine Learning techniques and algorithms.
5	Apply various algorithms to different datasets.

Sr. No.	Practical Description
Concern faculty member should suitably frame at least Eight laboratory assignments from the following list.	
1	Load and explore a dataset (e.g., Iris, MNIST) using Pandas, calculating summary statistics.
2	Split data into train/validation/test sets (70-15-15) and save using train_test_split.
3	Implement k-NN from scratch in Python to classify Iris flowers, varying k to observe accuracy changes.

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4	Train a linear regression model on Boston Housing data, visualizing residuals vs. predictions.
5	Build a logistic regression classifier for spam detection, evaluating with precision-recall curves.
6	Optimize SVM hyperparameters (C, kernel) via grid search on a synthetic 2D dataset.
7	Compute confusion matrices and F1-scores for a diabetes prediction model (Pima dataset).
8	Plot ROC curves for a binary classifier, comparing AUC scores for different algorithms.
9	Cluster customer data using k-means (scikit-learn), analyzing silhouette scores for optimal k.
10	Implement hierarchical clustering (agglomerative) on gene expression data, visualizing dendrograms.
11	Reduce 30D breast cancer features to 2D using PCA and plot decision boundaries.
12	Train a Gaussian Mixture Model (GMM) to identify subpopulations in height-weight data.
13	Preprocess text data (TF-IDF) and classify movie reviews using Naive Bayes.
14	Fine-tune a pre-trained CNN (VGG16) on CIFAR-10, freezing early layers.
15	Compare LDA vs. PCA for facial recognition on the Olivetti faces dataset.

Practical/Oral/Presentation:

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Dr. Ashutosh *S. K. Singh*